Q1. Solution:

Step 1

Assign **D1** → **Morning**

Constraint Check:

- D1 is not on Night ✓
- Shift available ✓

Accepted

Step 2

Assign **D2** → **Morning**

Morning already occupied ✗

Rejected

Try Afternoon

Accepted ✓

Step 3

Assign **D3** → **Morning**

D3 cannot work Morning ✗

Rejected

Try Afternoon

Already occupied ✗

Rejected

Try Night

Accepted ✓

Constraint Verification

Constraint	Status
D1 not Night	✓
D2 before D3	✓
One doctor per shift	✓

Constraint	Status
D3 not Morning	✓
One shift per doctor	✓

Final Schedule

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

Q2. Solution

Step 1: Start

- Start at **S (0,0)**.
- Add the start node to the queue.

Queue: [(0,0)]

Step 2: BFS Expansion

- Remove the first node from the queue.
- Visit all valid neighboring cells (Up, Down, Left, Right).
- Ignore obstacles (X) and visited cells.
- Add valid neighbors to the queue.

Step 3: Continue Search

- Repeat the process until the Goal (G) is reached.

Optimal Path

(0,0)

→ (1,0)

→ (2,0)

→ (2,1)

→ (2,2)

→ (3,2)

→ (4,2)

→ (4,3)

→ (4,4)

Total Cost

- Number of moves = **8**
- Cost per move = **1**

Total Cost = 8

Manhattan Distance Formula

$$h(n) = | x_1 - x_2 | + | y_1 - y_2 |$$

Q3. Solution

(a) Constraint Analysis

- The robot moves only Up, Down, Left, and Right.
- It cannot pass through obstacles (X).
- Each move costs 1.
- Entering a Risk Zone (R) adds 2 extra cost.
- The robot updates its path based on nearby information.
- The objective is to reach Goal (G) with the minimum total cost.

(b) Solution Approach (Uniform Cost Search)

1. Start from S.
2. Store nodes in a priority queue.
3. Expand the node with the lowest cost.
4. Skip obstacles.
5. Add 1 for each move and +2 for risk zones.
6. Repeat until the robot reaches G.

(c) AI Concepts

- **Intelligent Agent:** The robot senses the environment and makes decisions.
- **Rationality:** It always chooses the path with the lowest cost.
- **Environment Type:** Partially observable, dynamic, sequential, and discrete.

(d) Justification

Uniform Cost Search is suitable because it:

- Finds the minimum-cost path.
- Considers both movement and risk costs.
- Avoids obstacles.
- Produces the optimal solution.

Final Result

Optimal Path:

S → (1,0) → (2,0) → (2,1) → (3,1) → (3,2) → (4,2) → (4,3) → G

Total Cost = 8

Conclusion

The Uniform Cost Search (UCS) algorithm successfully guides the rescue robot to the survivor while minimizing the total travel cost and satisfying all the given constraints.